Decode lab IOT

Project 3 : Cloud Connected Security Node

Project Objective:

The purpose of this project is to build a cloud- connected security node

usingIoT technology. The system detects distance using an ultrasonic

sensor andsends live telem

etry data to a cloud dashboard through W i-Fi

using MQTT orHTTP protocol.

Components Required:

•Arduino Uno

•HC-SR04 Ultrasonic Sensor

•LED

•Buzzer

•Breadboard

•Jumper Wires

•USB Cable

Working Principle:

•Ultrasonic sensor detects object distance.

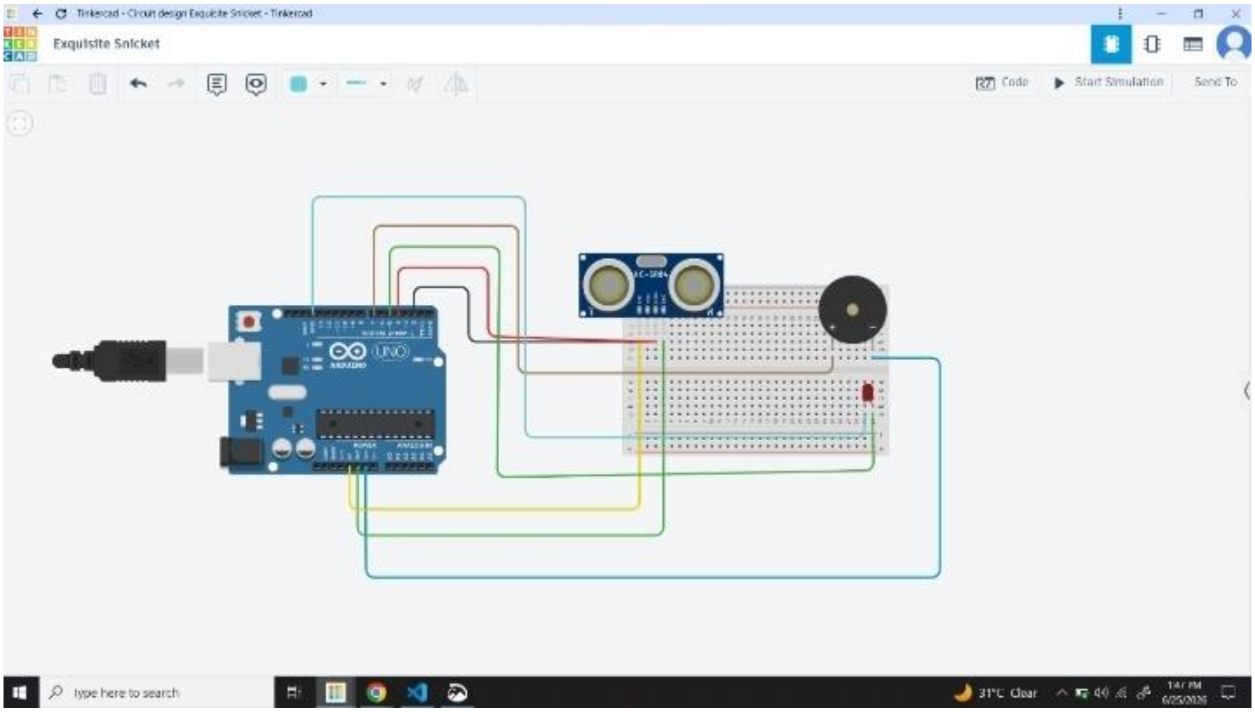
•If distance becomes less than the set limit:

°LED turns ON

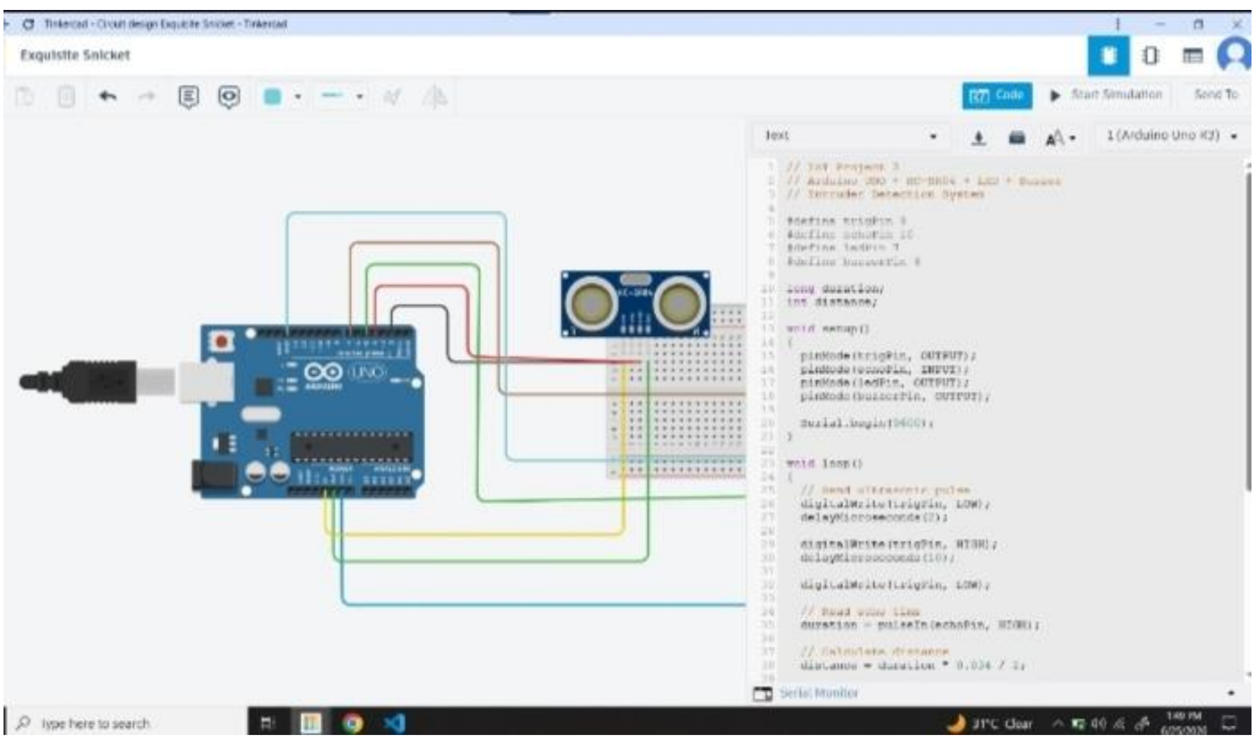
°Buzzer starts alarm

•This project works perfectly on tinkercad.com without ESP8266.

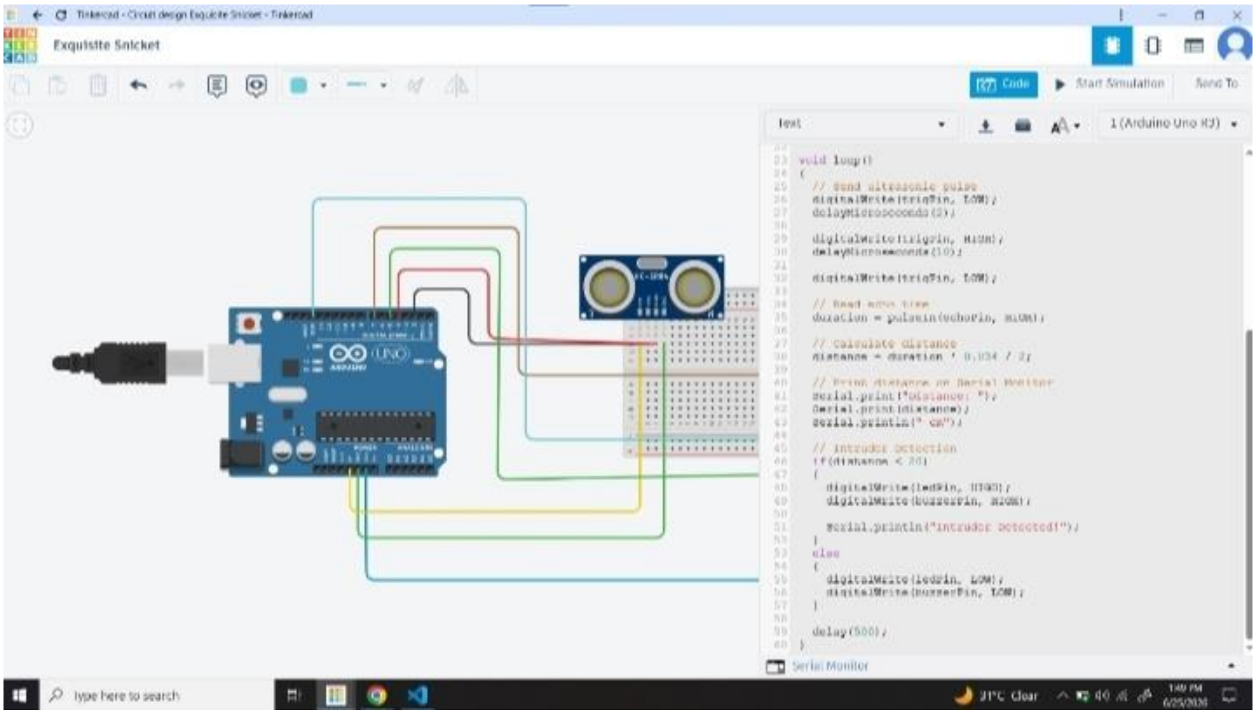
Circuit Diagram:

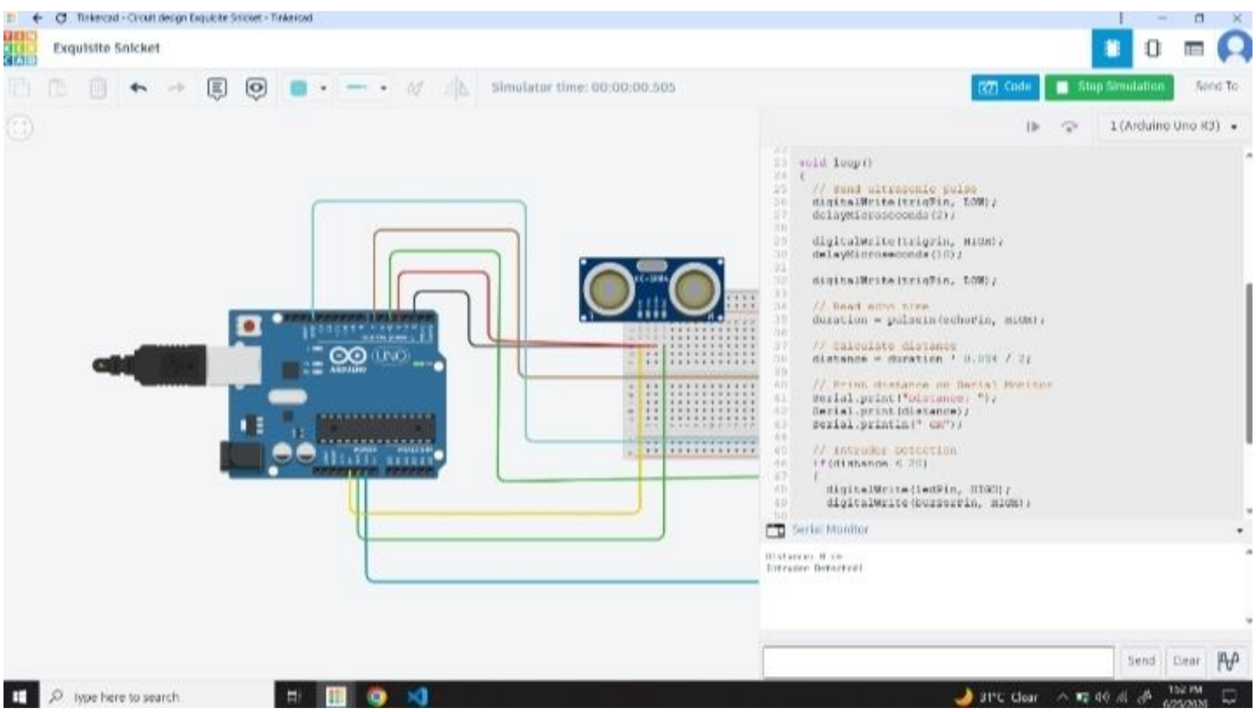


Exquisite Snicket
Code
Start Simulation
Send To
Type here to search



Exquisite Snicket
Code
Start Simulation
Send To
Text
1 (Arduino Uno R3)
long duration;
void setup()
pinMode(trigPin, OUTPUT);
pinMode(ledPin, OUTPUT);
void loop()
digitalWrite(trigPin, HIGH);
duration = pulseIn(echoPin, HIGH);
Serial Monitor
Type here to search





Circuit Connections:

1.HC- SR04 VCC 5V

2.HC- SR04 GND GND

3.HC- SR04 Trig D5

4.HC- SR04 Echo D6

Cloud Dashboard Setup (Blynk):

5.Create a Blynk account.

6.Create a new template and device.

7.Copy Template ID and Auth Token.

8.Add Gauge widget connected to V0.

9.Upload the code to ESP8266.

10. Open dashboard to monitor live distance values.

Arduino Code:

```
// IoT Project 3
```

```
// Arduino UNO + HC-SR04 + LED +
Buzzer
// Intruder Detection System

#define trigPin 9
#define echoPin 10
#define ledPin 7
#define buzzerPin 6

long duration;
int distance;

void setup()
{
```

```
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  Serial.begin(9600);
}

void loop()
{
  // Send ultrasonic pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
```

```
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);

digitalWrite(trigPin, LOW);

// Read echo time
duration = pulseIn(echoPin, HIGH);

// Calculate distance
distance = duration * 0.034 / 2;

// Print distance on Serial Monitor
```

```
Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");

// Intruder Detection
if(distance < 20)
{
  digitalWrite(ledPin, HIGH);
  digitalWrite(buzzerPin, HIGH);

  Serial.println("Intruder Detected!");
}
else
```

```
 {
   digitalWrite(ledPin, LOW);
   digitalWrite(buzzerPin, LOW);
 }

 delay(500);
}
```

Output:

The dashboard will display live distance values. W hen an object or intruder

comes near the sensor, the cloud dashboard updates instantly and sends an alert notification.

Advantages:

•Real- time cloud monitoring

•Remote access through internet

•Low- cost IoT security system

•Easy integration with mobile dashboards

Conclusion:

This project successfully demonstrates IoT telemetry using W i-Fi

Communication, cloud dashboard integration, and live sensor monitoring. It

Fulfills all major requirements of the DecodeLabs Project 3 assignment.